NO NONCONSTANT MORPHISM FROM THE PROJECTIVE LINE TO A GROUP SPACE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers within this span itself.

Allowed standard prerequisites (editorial).

- Product-power factorization for P1-powers. lemma-quotient-power-P1 (same chapter, NOT carded, declared external). Makes the n-fold power map finite onto its image.
- Dimension of a group space over a field. lemma-group-over-field-locally-finite-type-dimension (same chapter, NOT carded, declared external). dim(G) is finite for G locally of finite type.

Source reference. Original lemma 06FE: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1160–1160. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AEM: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1426–1426. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let G be a separated group algebraic space locally of finite type over k. There does not exist a nonconstant morphism $f : \mathbf{P}^1_k \to G$ over* $\mathrm{Spec}(k)$.

Proof. Assume f is nonconstant. Consider the morphisms

$$\mathbf{P}^1_k \times_{\mathrm{Spec}(k)} \ldots \times_{\mathrm{Spec}(k)} \mathbf{P}^1_k \longrightarrow G, \quad (t_1, \ldots, t_n) \longmapsto f(g_1) \ldots f(g_n)$$

where on the right hand side we use multiplication in the group. By Lemma 0AEM and the assumption that f is nonconstant this morphism is finite onto its image. Hence $\dim(G) \geq n$ for all n, which is impossible by Lemma 06FE and the fact that G is locally of finite type over k. □

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.